

Uniform bounds for the transfer moments

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Abstract

The constants of the density-transfer theorem (unit 83), the weighted remainder moment M_H and the tail moments T_i , are bounded explicitly by the log-weighted kernel moments $\Lambda_{\gamma,j}(a') = \int_0^\infty s^\gamma (1 + |\log s|)^j e^{-\beta s^2 + \beta a' s} ds$ under the polynomial-exponential bound $|c(s)| \leq e^{\beta s a'} (A + Bs + Ds^2)$. Since $\Lambda_{\gamma,j}$ is monotone in a' , the bounds are uniform over face parameters $|a| \leq L$, the input for the second-order theorem with noncritical coordinates. The third GPT-6 Astra consult ([tide-log/gpt6_bigpicture_v3.md](#)) fixes the architecture of the all-order theory: finite compatible jets with moment-integrable remainders as the engine, analyticity only as a provider of jets and of coefficient-series representations. Zero `sorry/axiom`.

1 The things formalised

```
noncomputable def logKernelMoment ( a : ℝ ) ( j : ℕ ) : ℝ :=
  s in Ioi ( 0 : ℝ ), ( 1 + |Real.log s| ) ^ j * ( s ^ a * quadKernel a s )
theorem logKernelMoment_mono ( a : ℝ ) ( h : 0 < a ) ( h' : -1 < a ) ( j : ℕ ) { a' a'' : ℝ }
  ( haa : a < a' ) : logKernelMoment a j < logKernelMoment a' j
theorem tailMoment_le_of_bound ( a' a A B D : ℝ ) ( h : 0 < a ) ( ha : -1 < a - 1 )
  ( j : ℕ ) ( c : ℝ → ℝ ) ( hc : Measurable c )
  ( hbound : s, 0 < s → |c s| ≤ Real.exp ( s * a' ) * ( A + B * s + D * s ^ 2 ) ) :
  tailMoment a j c ≤ A * logKernelMoment a' ( a - 1 ) j + B * logKernelMoment a' a j
  + D * logKernelMoment a' ( a + 1 ) j
theorem envMoment_le_of_bound ... :
  envMoment a J H ≤ A * logKernelMoment a' ( a - 1 ) J + B * logKernelMoment a' a J
  + D * logKernelMoment a' ( a + 1 ) J
```

2 Roadblocks and resolutions

- The `Pi.add` form of `Integrable.add` again defeats `← integral_add`; name the sums with single-lambda type ascriptions.

3 Outcome

Astra #3 ranking: (1) signed leading limit and the correct nonnegative-face variant (done except the “not identically zero” phrasing); (2) abstract all-order 1D kernel-jet transfer followed by the polynomial jet recursion $E_n = \frac{\beta s}{n} \sum_i i x_i E_{n-i}$, $P_j = \sum_\ell y_\ell E_{j-\ell}$, $c_j = k^{-1} \sum_\ell p_{j,\ell} S_{q_j+\ell}(a)$; (3) $d = 2$ block with noncritical face parameters under a strict gap; (4) weighted finite-part axis theorem and the cutoff-indexed density expansion; (5) a narrow tree identification plus the L^1 interchange lemma. Next: target (2).